

Work Stream 5: Incompressibility vs. Divergence-Free Vectors

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1 Introduction

A common point of confusion in interdisciplinary fluid dynamics, raised insightfully by u/brokebeany, is the distinction between a fluid being *divergence-free* and a fluid being *incompressible*.

2 The Mathematical vs. Physical Gap

In the OpenAI Lean 4 proof, the constructed vector field $u(x, t)$ satisfies:

$$\nabla \cdot u = 0 \quad \text{for all } t \in [0, 1) \quad (1)$$

Mathematically, this means the PDE is perfectly satisfied. However, in physics, the equation $\nabla \cdot u = 0$ is derived from the continuity equation $\partial_t \rho + \nabla \cdot (\rho u) = 0$ by assuming that density ρ is strictly constant.

This assumption is an asymptotic limit that only holds when the Mach number $Ma = |u|/c_s \ll 1$. The precise statement uses the bulk modulus $K = \rho c_s^2$: a dynamic pressure change $\Delta p \sim \rho u^2$ produces a density change

$$\frac{\Delta \rho}{\rho} = \frac{\Delta p}{K} \sim \frac{\rho u^2}{\rho c_s^2} = Ma^2, \quad (2)$$

so “constant density” means $Ma^2 \ll 1$. When the fluid in the OpenAI singularity breaches $Ma = 0.3$ (at roughly $t \approx 6.7$ picoseconds before blow-up, i.e. physical time $t = T\tau$ with $T = \ell_0^2/\nu = 100$ s for the $\ell_0 = 1$ cm water example — not the dimensionless τ read directly as seconds), physical density variations scale as $\Delta \rho/\rho \sim Ma^2 \sim 0.1$.

This number is not an artefact of the 1 cm example: because the collapsing core keeps a radial Reynolds number of order one, $u \simeq \sqrt{\nu/t}$ up to a factor $(\ell_0^2/\nu t)^h$ with $h = 1/200$, so the time at which $Ma = 0.3$ is reached, $t \simeq \nu/(0.3 c_s)^2 \approx 5$ ps, changes by less than 30% over six decades of ℓ_0 .

Therefore, while the *vector field* remains mathematically divergence-free, the *physical fluid* it is supposed to represent becomes compressible. The equation stops matching reality.

3 Acknowledgments

We thank u/brokebeany for prompting this critical clarification on the ontology of fluid mechanics.